GATE

# Computer Science & IT

## Computer Networks

## IPv4 Header

DPP: 1

**Q1** Consider IPv4 packet starting bits are "01000111", calculate header size (in bytes)?
(A) 5 (B) 7
(C) 20 (D) 28

**Q2** An IPv4 datagram of size 4000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 500 bytes. Assume that the size of the IP header is 20 bytes. The size of the last fragment (in bytes) is _____.

**Q3** Consider TCP segment of size 2000 bytes is passed to IP for delivery. MTU(maximum transmission unit) for source network is 300 bytes and IPv4 header size is 20 bytes then calculate offset value of the last fragments?

**Q4** Host A sends an IPv4 datagram to host B. When the datagram reaches at host B, which of the following IP header field(s) may be different from that of the original datagram?
(A) TTL
(B) Checksum
(C) Protocol Type
(D) Identification Number

**Q5** Consider Initial bits of an IPv4 packet header (in hexadecimal) are "0x4C". How many words of options are being carried by this packet in its header?

**Q6** The TTL (time-to-live) field of an IPv4 datagram is 3, how many hops can this packet travel before being dropped?
(A) 1 hops (B) 2 hops
(C) 3 hops (D) 4 hops

**Q7** In the TCP/IP protocol suite, which one of the following is NOT part of the IP header?
(A) Source IP Address
(B) Destination IP Address
(C) Source Port Number
(D) Destination Port Number

**Q8** Consider TCP segment of size 1540 bytes is passed to IP for delivery. MTU (maximum transmission unit) for source network is 300 bytes and IPv4 header size is 40 bytes then calculate total number of fragments?

Android App | iOS App | PW Website

GATE

# Answer Key

| | | | |
|---|---|---|---|
| Q1 | (D) | Q5 | 7 |
| Q2 | 160 | Q6 | (C) |
| Q3 | 245 | Q7 | (C, D) |
| Q4 | (A, B) | Q8 | 6 |

Android App | iOS App | PW Website

GATE

# Hints & Solutions

**Q1 Text Solution:**

HLEN= 0111= 7

Header Size= 7*4= 28 byte

**Q2 Text Solution:**

Old Payload size= 4000-20= 3980 byte

New Datagram:

MTU= 500 byte

Header size= 20 byte

New Payload Size= 500-20= 480 byte

SO, Number of Fragments= ceil ( 3980 / 480 ) = 9

So , Last fragment size= [ Old payload size - (N-1)*New Payload size ]= 3980 - 8*480 = 140 bytes

So, Size of the last fragment= Header size + Payload = 20 + 140 = 160 bytes.

**Q3 Text Solution:**

TCP segment size= 2000 byte.

MTU= 300 byte

Payload= 300 - 20= 280 byte

N= Number of segment= ceil (2000 / 280) = 8

offset of last fragment= [ (N - 1) * Payload Size ] / 8= 7*280 / 8 = 245

**Q4 Text Solution:**

Assigned by source only does not during routing.

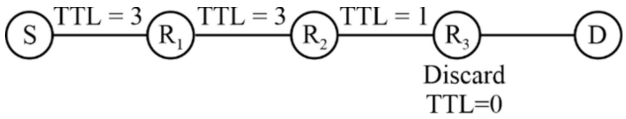
**Q5 Text Solution:**

VER= 0100

HLEN= 1100 = 12

So size of options= (HLEN - 5) words = 12-5 = 7

**Q6 Text Solution:**




**Q7 Text Solution:**

- Source Port Number
- Destination Port Number

TCP or UDP packet

**Q8 Text Solution:**

TCP seg = 1540 byte

Header Size= 40 byte

MTU = 300 bye

Payload size= ( MTU- Header Size)= 300-20= 280 byte[ not multiple of 8]

so we have to adjust and take value less than and near about 280 which is multiple of 8.

we take= 256 byte

256(1st ) 256(2nd ) 256( 3rd ) 256(4th ) 256(5th ) 260(6th )

Android App | iOS App | PW Website